

I lead a research team at Apodex working on large language models for mathematics and verification. AI systems generate code and proofs faster than anyone can check them; as generation gets cheaper, verification becomes the bottleneck. I have worked on machine learning for formal mathematics since my PhD.

EXPERIENCE

Apodex, Redwood City, CA
Lead Scientist

Mar 2026 – Present

- Lead a team of five researchers on the mathematical capabilities of LLMs across the training stack: pre-training data that aims to capture all the math on the web, synthetic mid-training data distilled from research papers, and post-training pushing toward research-level mathematics.
- Build verifiers that help agents complete long-horizon tasks, applied to AI's own training pipelines, from the correctness of GPU kernels to the quality of training data.

Meta Fundamental AI Research (FAIR), New York, NY
Research Scientist

Jun 2024 – Dec 2025

- Advised Goedel-Prover and Goedel-Prover-V2 (COLM 2025, ICLR 2026), theorem provers that set the state of the art among open models at release, outperforming provers $20\times$ their size; V2 is downloaded about 10,000 times a month.
- Co-developed ProofOptimizer (ICLR 2026), a model that simplifies machine-generated proofs without human demonstrations; collaborated on verified code generation (Verina, ICLR 2026; early adopters include AWS and Harmonic).

California Institute of Technology, Pasadena, CA
Postdoctoral Fellow

Sep 2022 – May 2024

- Built **LeanDojo** (NeurIPS 2023), the first open-source infrastructure for theorem proving with LLMs in Lean, used by projects at Meta, ByteDance, and DeepSeek.
- Autoformalizing Euclidean Geometry (ICML 2024) enabled the first faithful formalization of Euclid's *Elements*, Book I, in a modern proof assistant.

EDUCATION

Princeton University — PhD in Computer Science

2022

University of Michigan — MS in Computer Science and Engineering

2018

Tsinghua University — B.Eng. in Computer Science and B.S. in Mathematics

2016

SELECTED PUBLICATIONS

Full list on [Google Scholar](#).

- **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models**
K. Yang, A. Swope, A. Gu, R. Chalamala, P. Song, S. Yu, S. Godil, R. Prenger, A. Anandkumar. NeurIPS 2023.
- **Formal Mathematical Reasoning: A New Frontier in AI**
K. Yang, G. Poesia, J. He, W. Li, K. Lauter, S. Chaudhuri, D. Song. ICML 2025; Communications of the ACM.
- **Goedel-Prover-V2: Scaling Formal Theorem Proving with Scaffolded Data Synthesis and Self-Correction**
Y. Lin, S. Tang, B. Lyu, Z. Yang, ..., K. Yang, ..., C. Jin. ICLR 2026.
- **Learning to Prove Theorems via Interacting with Proof Assistants**
K. Yang, J. Deng. ICML 2019.

OPEN SOURCE

Core developer of **LeanDojo**, **ReProver**, **Lean Copilot**, and **CoqGym** — open-source tooling for machine learning with the Lean and Coq proof assistants; 2,800+ GitHub stars across repositories.

SERVICE AND COMMUNITY

- Co-organizer of the MATH-AI workshop at NeurIPS 2023, 2025, 2026; Area Chair for ICML, NeurIPS, and ECCV.
- Invited talks at OpenAI, Google, Meta FAIR, the Simons Institute, UC Berkeley, Stanford, and CMU; guest co-instructor of *Advanced Large Language Model Agents* (UC Berkeley & MOOC, 2025).
- Mentored students and junior researchers who went on to PhD programs at CMU, Toronto, Cornell, CUHK, and UIUC, and to industry labs; work covered in *Scientific American* (2024).